PROBLEM 33

Two Repelling Particles on a Rotating Parabola

Original statement · Problems.pdf, p. 21

Rotating Wire + Two Particles

Two heavy point particles, P_1 and P_2, of mass m each, slide without friction along a rigid wire parabola given by the equation $z = x^2/(2L)$, where L is a known length. The wire rotates about its axis with a fixed, known angular velocity ω. The two particles repel each other electrically with a force of magnitude $F_E = k/d^2$, where k is a known constant and d is the distance between the particles. We define the rotating reference frame $Oxyz$ attached to the parabola as shown in the figure below.

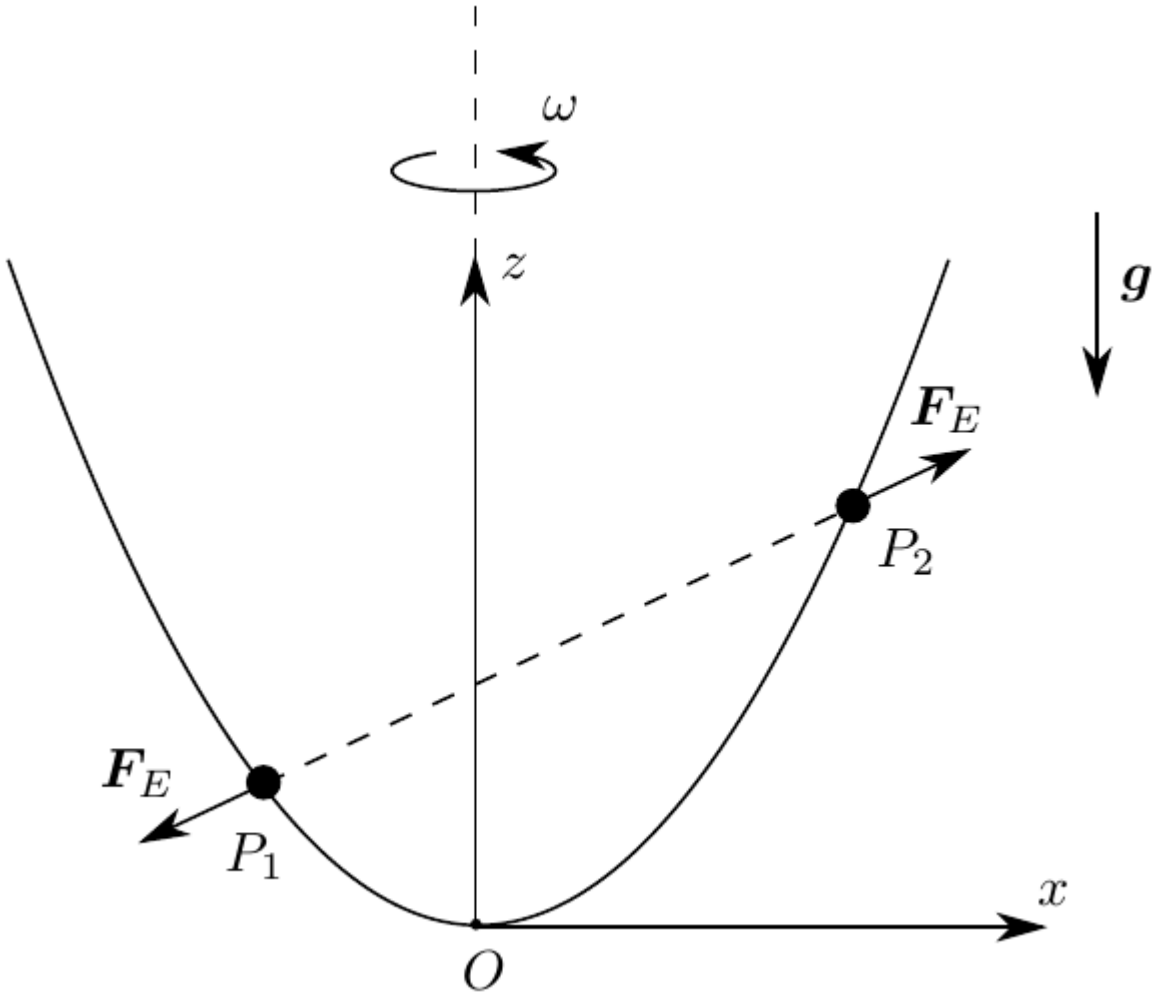


Figure: Parabola z = x²/(2L) in the plane Oxz rotating with angular velocity ω about Oz, particles P1 (left branch) and P2 (right branch) with the repulsive forces F_E along the line P1P2, gravity g downward (Problems.pdf, p. 21).

(a) Write down the position, velocity, and acceleration vectors of each particle relative to $Oxyz$ in a generic instant of time, as a function of x_1, x_2 (the position of P_1 and P_2 along the Ox axis) and their derivatives.

(b) Write down Newton's second law for each particle in the $Oxyz$ reference frame.

(c) If the rotation of the wire is maintained by a motor, how much torque must it produce in the Oz direction to keep ω constant as a function of the variables of the problem and their derivatives?

(d) Find out the equilibrium configurations of the system in the reference frame $Oxyz$ as a function of the parameters $\gamma = g/(\omega^2 L)$ and $\kappa = k/(m\omega^2 L^3)$.

Consider now that P_1 and P_2 are constrained to move symmetrically on each branch of the parabola, and that $\gamma = \kappa = 2$. Initially, $x_1(0) = L$ and $v_1(0) = L\omega$ (with $\dot{x}_1(0) > 0$).

(e) Write down the equation that gives the minimum and maximum values of x_1 occurring in this case. *Hint: this equation is a third-order polynomial in x_1.*

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** two particles P_1, P_2 of mass m slide on a smooth parabolic wire $z = x^2/(2L)$ that spins about its vertical axis Oz at constant ω. They repel each other with $F_E = k/d^2$ along the line that joins them, and both are heavy.
- **Why the rotating frame $Oxyz$.** In $Oxyz$ the wire is fixed, so each particle is located by a single coordinate x_i and its relative kinematics is that of a point on a fixed plane curve. The frame rotates uniformly about a fixed axis through O, so the inertia forces reduce to the centrifugal and Coriolis forces (Slide 28), and the centrifugal force has a potential (Slide 30).
- **Hypotheses:**
  1. The frame $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, with O the vertex of the parabola and $\mathbf{k}_0$ vertical upward, is inertial.
  2. The wire is smooth and bilateral (each particle is a bead): its reaction on P_i is perpendicular to the wire, with components in the plane of the wire and along $\mathbf{j}_1$.
  3. ω is kept constant by the motor; the axis bearings exert no torque about Oz.
  4. Electric repulsion is a central internal force: equal and opposite on the two particles, along $\overrightarrow{P_2P_1}$.
- **Degrees of freedom:** two, the abscissas x_1 and x_2 in $Oxyz$. In part (e) the symmetry constraint $x_2 = -x_1$ leaves one.

Symbol	Meaning	SI unit
m	mass of each particle	kg
L	parameter of the parabola $z = x^2/(2L)$ (the radius of curvature at the vertex)	m
ω	constant angular velocity of the wire	rad/s
g	gravitational acceleration	m/s^2
k	electric repulsion constant, $F_E = k/d^2$	N·m^2
$\gamma = g/(\omega^2 L)$	weight versus centrifugal force	1
$\kappa = k/(m\omega^2 L^3)$	repulsion versus centrifugal force	1

PHASE 2

Reference Frames & Vector Bases

- **Frames.** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ inertial; $S_1 = Oxyz = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ attached to the wire, with the wire in the plane Oxz and $\mathbf{k}_1 = \mathbf{k}_0$ along the axis. The two frames share the origin O, which lies on the axis and is fixed.
- **Change of basis** (rotation of angle ωt about $\mathbf{k}_0$, with $\mathbf{i}_1 = \mathbf{i}_0$ at $t = 0$; columns are $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$):

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1$$

The columns are unit vectors and mutually orthogonal, so $[{}_0R_1][{}_0R_1]^T = I$.

- **Motion of S_1:** $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$; $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$; $\boldsymbol{\alpha}_{10} = \mathbf{0}$ because ω is constant.
- **Local geometry of the wire at P_i.** A tangent vector is $\dfrac{d}{dx}\left(x\,\mathbf{i}_1 + \dfrac{x^2}{2L}\mathbf{k}_1\right) = \mathbf{i}_1 + \dfrac{x}{L}\mathbf{k}_1$. With $s_i = \sqrt{1 + x_i^2/L^2}$, the unit tangent and the in-plane unit normal (towards the concave side) are

$$\mathbf{t}_i = \frac{1}{s_i}\left(\mathbf{i}_1 + \frac{x_i}{L}\mathbf{k}_1\right), \qquad \mathbf{n}_i = \frac{1}{s_i}\left(-\frac{x_i}{L}\mathbf{i}_1 + \mathbf{k}_1\right)$$

They are orthogonal, $\mathbf{t}_i \cdot \mathbf{n}_i = \dfrac{1}{s_i^2}\left(-\dfrac{x_i}{L} + \dfrac{x_i}{L}\right) = 0$, and, using $\mathbf{i}_1 \times \mathbf{k}_1 = -\mathbf{j}_1$ and $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$,

$$\mathbf{t}_i \times \mathbf{n}_i = \frac{1}{s_i^2}\left[\mathbf{i}_1 \times \mathbf{k}_1 - \frac{x_i^2}{L^2}\,\mathbf{k}_1 \times \mathbf{i}_1\right] = \frac{1}{s_i^2}\left[-\mathbf{j}_1 - \frac{x_i^2}{L^2}\mathbf{j}_1\right] = -\mathbf{j}_1$$

so $\{\mathbf{t}_i, \mathbf{n}_i, -\mathbf{j}_1\}$ is a right-handed orthonormal triad attached to the wire at P_i. The reaction of the smooth wire lies in the plane spanned by $\mathbf{n}_i$ and $\mathbf{j}_1$.

PHASE 3

Derivation

3.1 Kinematics

(a) Relative position, velocity and acceleration

Pedagogical Justification: Relative kinematics is computed in S_1, where $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ are constant, so only the scalar components are differentiated. Each $z_i = x_i^2/(2L)$ depends on time through $x_i(t)$, so the chain rule is needed.

For $i = 1, 2$, the position on the wire is

$$\mathbf{r}_1^{P_i} = x_i\,\mathbf{i}_1 + \frac{x_i^2}{2L}\,\mathbf{k}_1$$

Chain rule: $\dfrac{d}{dt}\left(\dfrac{x_i^2}{2L}\right) = \dfrac{2x_i}{2L}\,\dot{x}_i = \dfrac{x_i\dot{x}_i}{L}$. Hence

$$\boxed{\mathbf{v}_1^{P_i} = \dot{x}_i\,\mathbf{i}_1 + \frac{x_i\dot{x}_i}{L}\,\mathbf{k}_1}$$

Product rule on $x_i\dot{x}_i$: $\dfrac{d}{dt}(x_i\dot{x}_i) = \dot{x}_i\dot{x}_i + x_i\ddot{x}_i$. Hence

$$\boxed{\mathbf{a}_1^{P_i} = \ddot{x}_i\,\mathbf{i}_1 + \frac{\dot{x}_i^2 + x_i\ddot{x}_i}{L}\,\mathbf{k}_1}$$

Note $\mathbf{v}_1^{P_i} = s_i \dot{x}_i \, \mathbf{t}_i$, tangent to the wire, as it must be.

3.2 Forces

(b) Newton's second law in $Oxyz$

> **Pedagogical Justification:** $Oxyz$ is not inertial; Newton's law holds there if the inertia forces are added, $m\mathbf{a}_1^{P_i} = \mathbf{F} + \mathbf{F}_i$ (Notes Eqs. 4.62–4.63). Projecting on the tangent $\mathbf{t}_i$ eliminates the unknown reaction and gives the equation of motion; the other two projections give the reaction.

Actual forces on P_i:

- Weight: $-mg\,\mathbf{k}_1$.
- Reaction of the smooth wire: $\mathbf{N}_i = N_{in}\,\mathbf{n}_i + N_{iy}\,\mathbf{j}_1$ (so $\mathbf{N}_i \cdot \mathbf{t}_i = 0$).
- Electric repulsion. With $\Delta x = x_1 - x_2$ and $\Delta z = \dfrac{x_1^2 - x_2^2}{2L} = \Delta x \, \dfrac{x_1 + x_2}{2L}$,

$$\overrightarrow{P_2P_1} = \Delta x \left[\mathbf{i}_1 + \frac{x_1 + x_2}{2L}\mathbf{k}_1\right], \qquad d^2 = |\overrightarrow{P_2P_1}|^2 = \Delta x^2 \left[1 + \frac{(x_1 + x_2)^2}{4L^2}\right]$$

The force on P_1 has magnitude k/d^2 and the direction of $\overrightarrow{P_2P_1}/d$, and the force on P_2 is opposite:

$$\mathbf{F}_{E1} = \frac{k}{d^3}\,\Delta x \left[\mathbf{i}_1 + \frac{x_1 + x_2}{2L}\mathbf{k}_1\right] = -\mathbf{F}_{E2}$$

Inertia forces ($\mathbf{a}_0^O = \mathbf{0}$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$, so $\mathbf{F}_A = \mathbf{F}_E^{\text{Euler}} = \mathbf{0}$):

- Centrifugal: $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^{P_i} = \omega\mathbf{k}_1 \times \left(x_i\mathbf{i}_1 + \frac{x_i^2}{2L}\mathbf{k}_1\right) = \omega x_i\,\mathbf{j}_1$, then $\boldsymbol{\omega}_{10} \times \omega x_i \mathbf{j}_1 = \omega^2 x_i\,(\mathbf{k}_1 \times \mathbf{j}_1) = -\omega^2 x_i\,\mathbf{i}_1$, so $\mathbf{F}_{C,i} = m\omega^2 x_i\,\mathbf{i}_1$ (outward from the axis, Slide 30).
- Coriolis: $\boldsymbol{\omega}_{10} \times \mathbf{v}_1^{P_i} = \omega\mathbf{k}_1 \times \left(\dot{x}_i\mathbf{i}_1 + \frac{x_i\dot{x}_i}{L}\mathbf{k}_1\right) = \omega\dot{x}_i\,\mathbf{j}_1$, so $\mathbf{F}_{COR,i} = -2m\omega\dot{x}_i\,\mathbf{j}_1$.

3.3 Equations of Motion

(b, continued) Newton's equations

Newton's second law for each particle (vector form):

$$m\left[\ddot{x}_i\,\mathbf{i}_1+\frac{\dot{x}_i^2+x_i\ddot{x}_i}{L}\,\mathbf{k}_1\right]=-mg\,\mathbf{k}_1+N_{in}\,\mathbf{n}_i+N_{iy}\,\mathbf{j}_1+\mathbf{F}_{Ei}+m\omega^2x_i\,\mathbf{i}_1-2m\omega\dot{x}_i\,\mathbf{j}_1,\qquad i=1,2$$

Projection on $s_i\mathbf{t}_i=\mathbf{i}_1+\dfrac{x_i}{L}\mathbf{k}_1$ (no reaction). Left side: $m\left[\ddot{x}_i+\dfrac{x_i}{L}\cdot\dfrac{\dot{x}_i^2+x_i\ddot{x}_i}{L}\right]=m\left[\left(1+\dfrac{x_i^2}{L^2}\right)\ddot{x}_i+\dfrac{x_i\dot{x}_i^2}{L^2}\right]$. Right side: weight $-mg\,x_i/L$; centrifugal $m\omega^2x_i$; Coriolis 0; electric, for P_1,

$$\mathbf{F}_{E1}\cdot\left(\mathbf{i}_1+\frac{x_1}{L}\mathbf{k}_1\right)=\frac{k}{d^3}\,(x_1-x_2)\left[1+\frac{x_1(x_1+x_2)}{2L^2}\right]$$

and for P_2 the same with the indices exchanged (because $\mathbf{F}_{E2}=-\mathbf{F}_{E1}$ and $-\Delta x=x_2-x_1$). The **equations of motion** are

$$\boxed{m\left[\left(1+\frac{x_1^2}{L^2}\right)\ddot{x}_1+\frac{x_1\dot{x}_1^2}{L^2}\right]=m\left(\omega^2-\frac{g}{L}\right)x_1+\frac{k}{d^3}\,(x_1-x_2)\left[1+\frac{x_1(x_1+x_2)}{2L^2}\right]}$$

$$\boxed{m\left[\left(1+\frac{x_2^2}{L^2}\right)\ddot{x}_2+\frac{x_2\dot{x}_2^2}{L^2}\right]=m\left(\omega^2-\frac{g}{L}\right)x_2+\frac{k}{d^3}\,(x_2-x_1)\left[1+\frac{x_2(x_1+x_2)}{2L^2}\right]}$$

with $d^3=|x_1-x_2|^3\left[1+\dfrac{(x_1+x_2)^2}{4L^2}\right]^{3/2}$.

Projection on $\mathbf{j}_1$: $0=N_{iy}-2m\omega\dot{x}_i$, so $\boxed{N_{iy}=2m\omega\dot{x}_i}$: the wire pushes each bead sideways to cancel the Coriolis force.

Projection on $s_i\mathbf{n}_i=-\dfrac{x_i}{L}\mathbf{i}_1+\mathbf{k}_1$. Left side: $m\left[-\dfrac{x_i\ddot{x}_i}{L}+\dfrac{\dot{x}_i^2+x_i\ddot{x}_i}{L}\right]=\dfrac{m\dot{x}_i^2}{L}$. Right side: $N_{in}s_i-mg-\dfrac{m\omega^2x_i^2}{L}+\mathbf{F}_{Ei}\cdot(s_i\mathbf{n}_i)$, where for P_1

$$\mathbf{F}_{E1}\cdot\left(-\frac{x_1}{L}\mathbf{i}_1+\mathbf{k}_1\right)=\frac{k\,\Delta x}{d^3}\left[-\frac{x_1}{L}+\frac{x_1+x_2}{2L}\right]=\frac{k\,\Delta x}{d^3}\cdot\frac{x_2-x_1}{2L}=-\frac{k\,\Delta x^2}{2Ld^3}$$

and the same value is found for P_2. Therefore

$$\boxed{N_{in}=\frac{m}{s_i}\left[\frac{\dot{x}_i^2}{L}+g+\frac{\omega^2 x_i^2}{L}\right]+\frac{k\,(x_1-x_2)^2}{2L\,s_i\,d^3}},\qquad s_i=\sqrt{1+\frac{x_i^2}{L^2}}$$

3.4 Integration and Answers

(c) Motor torque that keeps ω constant

> **Pedagogical Justification:** The motor torque acts on the wire; the wire acts on the particles through the reactions $\mathbf{N}_i$, and the particles act back with $-\mathbf{N}_i$. The wire is a massless guide (if it had mass and were symmetric about Oz, rotating at constant ω, its own angular momentum about Oz would be constant; this remark goes beyond the particle units and does not change the result). Hence the moment about Oz of all the actions on it must vanish.

Balance of moments about Oz on the wire: $M+\sum_i\left[\mathbf{r}^{P_i}\times(-\mathbf{N}_i)\right]\cdot\mathbf{k}_1=0$. Expand the cross product with $\mathbf{r}^{P_i}=x_i\mathbf{i}_1+z_i\mathbf{k}_1$ and $\mathbf{N}_i=N_{ix}\mathbf{i}_1+N_{iy}\mathbf{j}_1+N_{iz}\mathbf{k}_1$; its $\mathbf{k}_1$ component is $x_iN_{iy}-0\cdot N_{ix}=x_iN_{iy}$. With $N_{iy}=2m\omega\dot{x}_i$ from part (b):

$$\boxed{M=x_1N_{1y}+x_2N_{2y}=2m\omega\left(x_1\dot{x}_1+x_2\dot{x}_2\right)}$$

Check with the angular momentum of the particles in S_0. By Notes Eq. 3.26, $\mathbf{v}_0^{P_i}=\mathbf{v}_1^{P_i}+\boldsymbol{\omega}_{10}\times\mathbf{r}_1^{P_i}=\dot{x}_i\mathbf{i}_1+\omega x_i\mathbf{j}_1+\frac{x_i\dot{x}_i}{L}\mathbf{k}_1$, whose $\mathbf{k}_1$-moment about O is $m\left[x_i(\omega x_i)-0\cdot\dot{x}_i\right]=m\omega x_i^2$. So $H_{Oz}=m\omega(x_1^2+x_2^2)$ and, by the product and chain rules, $\dot{H}_{Oz}=2m\omega(x_1\dot{x}_1+x_2\dot{x}_2)$. The weights are parallel to Oz and the electric pair is central ($(\mathbf{r}^{P_1}-\mathbf{r}^{P_2})\times\mathbf{F}_{E1}=\mathbf{0}$), so only the reactions have a moment about Oz, and that moment equals M. Both routes agree with the official key (Problems.pdf, p. 75).

(d) Equilibrium configurations in $Oxyz$

> **Pedagogical Justification:** A relative equilibrium has $\dot{x}_i = \ddot{x}_i = 0$; then the two equations of motion become algebraic. Writing them with $u_i = x_i/L$ and dividing by $m\omega^2 L$ shows that only γ and κ matter.

With $\dot{x}_i = \ddot{x}_i = 0$, $x_i = Lu_i$ and $d = LD$, $D = |u_1 - u_2|\sqrt{1 + (u_1+u_2)^2/4}$, dividing by $m\omega^2 L$:

$$f_1 \equiv (1-\gamma)\,u_1 + \frac{\kappa}{D^3}(u_1 - u_2)\left[1 + \frac{u_1(u_1+u_2)}{2}\right] = 0, \qquad f_2 \equiv (1-\gamma)\,u_2 + \frac{\kappa}{D^3}(u_2 - u_1)\left[1 + \frac{u_2(u_1+u_2)}{2}\right] = 0$$

Sum. With $S = u_1 + u_2$, the electric terms add to $\dfrac{\kappa}{D^3}(u_1 - u_2)\left[\dfrac{u_1 S}{2} - \dfrac{u_2 S}{2}\right] = \dfrac{\kappa\,(u_1-u_2)^2 S}{2D^3}$, so

$$f_1 + f_2 = S\left[(1-\gamma) + \frac{\kappa\,(u_1-u_2)^2}{2D^3}\right] = 0$$

Either $S = 0$ (symmetric configuration) or $\gamma - 1 = \dfrac{\kappa(u_1-u_2)^2}{2D^3} > 0$.

Case $\gamma < 1$. If $S = 0$, $u_2 = -u_1$, $D = 2|u_1|$ and $f_1 = u_1\left[(1-\gamma) + \dfrac{2\kappa}{8|u_1|^3}\right]$: both terms of the bracket are positive and $u_1 \neq 0$ (the particles cannot coincide), so $f_1 \neq 0$. If $S \neq 0$ the bracket of the sum is positive. **No equilibrium exists.** (Centrifugal force beats gravity everywhere and the repulsion only pushes the particles further out.)

Case $\gamma = 1$. Both alternatives require $\kappa = 0$. For $\kappa > 0$ there is no equilibrium; only without repulsion ($\kappa = 0$) is every configuration an equilibrium (gravity and centrifugal force balance along the wire at every point: $(\omega^2 - g/L)x_i = 0$).

Case $\gamma > 1$, symmetric ($x_2 = -x_1$). $f_1 = u_1\left[(1-\gamma) + \dfrac{\kappa}{4|u_1|^3}\right] = 0$ gives $|u_1|^3 = \dfrac{\kappa}{4(\gamma-1)}$:

$$\boxed{x_1 = -x_2, \qquad x_1^2 = \left[\frac{\kappa L^3}{4(\gamma-1)}\right]^{2/3}}$$

Case $\gamma > 1$, asymmetric ($S \neq 0$). Insert $\dfrac{\kappa}{D^3} = \dfrac{2(\gamma-1)}{(u_1-u_2)^2}$ from the sum into $f_1 = 0$ and divide by $(\gamma - 1)$:

$$-u_1 + \frac{2}{u_1-u_2}\left[1 + \frac{u_1(u_1+u_2)}{2}\right] = 0 \implies u_1(u_1-u_2) = 2 + u_1(u_1+u_2) \implies \boxed{x_1 x_2 = -L^2}$$

The particles are on opposite branches and the chord P_1P_2 passes through the focus $(0, L/2)$ of the parabola. Then $u_2 = -1/u_1$, $S = u_1 - 1/u_1$, $u_1 - u_2 = u_1 + 1/u_1$, and $1 + S^2/4 = (u_1 + 1/u_1)^2/4$, so $D^3 = |u_1 + 1/u_1|^6/8$ and the sum condition becomes

$$\gamma - 1 = \frac{4\kappa}{(u_1 + 1/u_1)^4} \implies \left|u_1 + \frac{1}{u_1}\right| = \left(\frac{4\kappa}{\gamma-1}\right)^{1/4}$$

Since $|u_1 + 1/u_1| \geq 2$, with equality only for $|u_1| = 1$ (which is the symmetric equilibrium), **asymmetric equilibria exist if and only if** $\kappa > 4(\gamma - 1)$, i.e. exactly when the symmetric equilibrium lies at $|x_1| > L$. They come in mirror pairs, $u_1 = \frac{1}{2}\left[q \pm \sqrt{q^2 - 4}\right]$ with $q = (4\kappa/(\gamma-1))^{1/4}$ and $u_2 = -1/u_1$ (plus the configurations with the labels exchanged). No equilibrium has both particles on the same branch.

(e) Symmetric motion with $\gamma = \kappa = 2$: turning points

> **Pedagogical Justification:** In $Oxyz$ the wire (and the symmetry constraint) is fixed, $\boldsymbol{\omega}_{10}$ is constant and the origin is fixed. The reactions and the Coriolis forces do no work in S_1 (Slide 31), and the weights, the electric pair and the centrifugal forces have potentials (Slides 29–30). The mechanical energy of the pair in S_1 is therefore conserved, and its turning points ($\dot{x}_1 = 0$) give the extreme values of x_1.

Put $x_1 = x > 0$, $x_2 = -x$ (the repulsion becomes infinite at $x = 0$, so x_1 stays positive). Then $d = 2x$ and, for each particle, $|\mathbf{v}_1^{P_i}|^2 = \dot{x}^2\left(1 + x^2/L^2\right)$.

- Kinetic energy: $T_1 = 2 \cdot \frac{1}{2} m\dot{x}^2\left(1 + \dfrac{x^2}{L^2}\right) = m\dot{x}^2\left(1 + \dfrac{x^2}{L^2}\right)$.
- Gravity: $2\,mg\,\dfrac{x^2}{2L} = \dfrac{mgx^2}{L}$. Centrifugal: $2\left(-\frac{1}{2}m\omega^2 x^2\right) = -m\omega^2 x^2$. Electric: $\dfrac{k}{d} = \dfrac{k}{2x}$ (its gradient gives the repulsion k/d^2).

$$E_1 = m\dot{x}^2\left(1+\frac{x^2}{L^2}\right) + m\left(\frac{g}{L}-\omega^2\right)x^2 + \frac{k}{2x} = \text{const}$$

Initial value. $|\mathbf{v}_1^{P_1}(0)| = L\omega$ at $x = L$ gives $2\dot{x}(0)^2 = L^2\omega^2$, so $T_1(0) = mL^2\omega^2$ and $E_1 = mL^2\omega^2 + m(g/L-\omega^2)L^2 + k/(2L)$. Dividing by $m\omega^2L^2$, with $u = x/L$:

$$\frac{E_1}{m\omega^2L^2} = \gamma + \frac{\kappa}{2} = 3 \qquad (\gamma = \kappa = 2)$$

Turning points ($\dot{x} = 0$): $(\gamma-1)u^2 + \dfrac{\kappa}{2u} = \gamma + \dfrac{\kappa}{2}$, i.e. $u^2 + \dfrac{1}{u} = 3$. Multiply by $u > 0$:

$$\boxed{\frac{x_1^3}{L^3} - 3\,\frac{x_1}{L} + 1 = 0}$$

Roots. With $u = 2\cos\theta$ and $4\cos^3\theta - 3\cos\theta = \cos 3\theta$: $8\cos^3\theta - 6\cos\theta + 1 = 2\cos 3\theta + 1 = 0$, so $\cos 3\theta = -\frac{1}{2}$ and $\theta = 2\pi/9,\ 4\pi/9,\ 8\pi/9$. The roots are $u = 2\cos(2\pi/9) = 1.532$, $2\cos(4\pi/9) = 0.347$ and $2\cos(8\pi/9) = -1.879$ (rejected, $x_1 > 0$). Since the starting point $u = 1$ lies between the two positive roots (there $u^2 + 1/u = 2 < 3$),

$$x_{1,\min} = 2L\cos\frac{4\pi}{9} \approx 0.347\,L, \qquad x_{1,\max} = 2L\cos\frac{2\pi}{9} \approx 1.532\,L$$

and x_1 oscillates between them. The key also lists $x_1^3/L^3 - 3x_1/L - 1 = 0$ for $x_1 < 0$: that is the mirror image (a start on the negative branch) and does not apply to $x_1(0) = L > 0$.

PHASE 4

Interpretation & Checks

- **Effective spring along the wire.** Gravity and the centrifugal force combine into $m(\omega^2 - g/L)x_i$ in the tangential equation: a restoring force if $\gamma > 1$ (gravity wins and pulls the beads to the vertex) and a repelling one if $\gamma < 1$. Equilibrium requires the repulsion to be balanced by this restoring force, hence the condition $\gamma > 1$.

- **Limits.** $\kappa \to 0$: the symmetric equilibrium collapses to $x_1 = x_2 = 0$ (no repulsion to separate the beads). $\gamma \to 1^+$: $x_1^2 \to \infty$, the restoring force vanishes and the beads are pushed to infinity.
- **Torque sign.** $M = \frac{d}{dt}\left[m\omega(x_1^2 + x_2^2)\right]$: when the beads move outward the moment of inertia of the system grows and the motor has to supply angular momentum ($M > 0$); when they move inward it must brake ($M < 0$).
- **Inertial check.** Differentiating $[{}_0R_1]\mathbf{r}_1^{P_i}$ twice reproduces $\mathbf{a}_1^{P_i} + \boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^{P_i}) + 2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^{P_i}$ (Notes Eq. 3.27), and $dE_1/dt = 0$ along the equations of motion of part (b) (both checked symbolically with sympy, together with the equilibria of part (d)).
- **Dimensions.** $[k/d^2] = \mathrm{N \cdot m^2/m^2} = \mathrm{N}$; $[m\omega^2 x] = \mathrm{kg \cdot s^{-2} \cdot m} = \mathrm{N}$; $[M] = [m\omega x\dot{x}] = \mathrm{kg \cdot s^{-1} \cdot m \cdot m/s} = \mathrm{N \cdot m}$; $[\gamma] = \dfrac{\mathrm{m/s^2}}{\mathrm{s^{-2}\,m}} = 1$; $[\kappa] = \dfrac{\mathrm{N \cdot m^2}}{\mathrm{kg \cdot s^{-2} \cdot m^3}} = 1$.
- **Check against the official key:** the key of `Problems.pdf` gives the motor torque $2m\omega\left(\dot{x}_1 x_1 + \dot{x}_2 x_2\right)$, the equilibrium cases ($\gamma < 1$: none; $\gamma = 1$: only for $\kappa = 0$; $\gamma > 1$: symmetric with $x_1 = -x_2$ and $x_1^2 = \left[\kappa L^3/\left(4(\gamma - 1)\right)\right]^{2/3}$, and asymmetric ones) and the turning-point equations $x_1^3/L^3 - 3x_1/L \pm 1 = 0$. They coincide with the results of (c), (d) and (e) (for $\gamma = \kappa = 2$ the turning-point condition $u^3 - 3u + 1 = 0$ is the positive-x_1 case).

Source: Problems.pdf (PDF page 21); key on PDF page 75